

Week 5 — Random Numbers & Rock-Paper-Scissors

Name: ____ Date: ____

randi rolls the dice. The **mapping** gives the numbers meaning. Every outcome gets an **equal chance**.

Today's words

Word	What it means
<code>randi([a b])</code>	a whole number from a to b — both ends included
<code>randi(n)</code>	shortcut for <code>randi([1 n])</code> — a number from 1 up to n
mapping	the if/elseif chain that gives each number a meaning (1 means rock...)
<code>strcmp(a, b)</code>	the word-matcher — "same word?" → 1 (yes) or 0 (no)
equally likely / fair	every outcome has the same chance — no player has an edge

1 • Prove the game is fair 🎲📄✂️

Fill every box with **W** (class wins), **L** (class loses), or **T** (tie). Remember: rock smashes scissors · paper wraps rock · scissors cut paper.

CLASS throws ↓ · COMPUTER throws →	rock	paper	scissors
rock			
paper			
scissors			

Count your letters: W: _ L: _ T: ____ → Is the game fair? **YES / NO**

2 • Which spinner is fair? 🎡

Three computer "spinners" — each rolls a die, then maps the number to a color or word. **Circle the fair one** (all outcomes equally likely):

- | | | | |
|------------------------------|-------------|--------------|-----------------------------|
| A) <code>r = randi(3)</code> | 1 means red | 2 means red | 3 means blue |
| B) <code>r = randi(3)</code> | 1 means red | 2 means blue | 3 means green |
| C) <code>r = randi(4)</code> | 1 means win | 2 means win | 3 means win 4 means lose |

In spinner A, the chance of red is _ **out of** . In spinner C, the chance of lose is _ **out of** .

3 • What's possible? 🎲

Circle **every** number that `randi([3 7])` could give:

0 1 2 3 4 5 6 7 8 9

🧠 Brain teaser (optional — take it home)

A sneaky computer plays rock-paper-scissors, but it picks its throw with this rigged mapping:

`r = randi(3)` 1 means rock 2 means rock 3 means paper

You get to pick ONE throw and stick with it. **Which throw gives you the best chance — and what is your chance of winning? Can you ever lose?** (Hint: list the 3 things the computer might do and what happens to your throw in each.) Bring your answer next week for a shout-out.